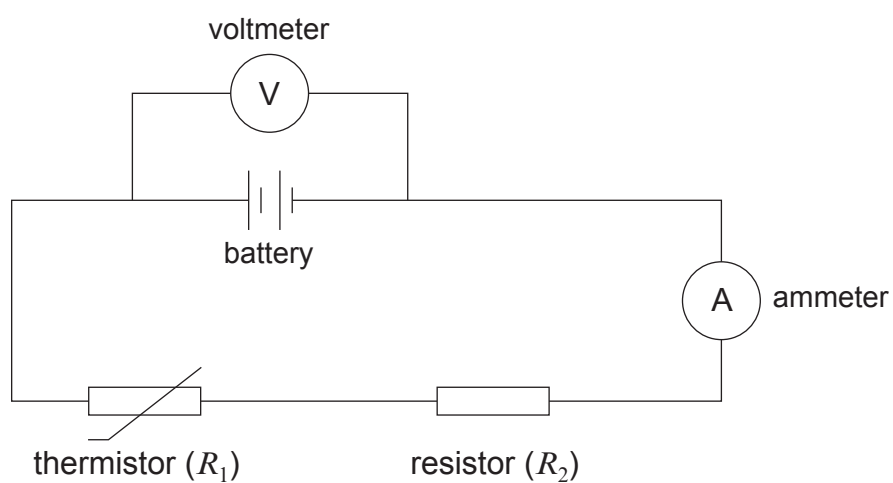


4. Students set up the temperature sensing circuit shown below.



(a) At a temperature of 20°C the resistance of the thermistor (R_1) is $300\ \Omega$.

The resistor (R_2) has a fixed resistance of $50\ \Omega$.

(i) Use the equation:

$$R = R_1 + R_2$$

to calculate the total resistance (R) of the circuit at 20°C .

[1]

total resistance = Ω



- (ii) A photograph of the voltmeter used in the circuit is shown below.



State the voltage shown on the voltmeter.

[1]

voltage = V

- (iii) Use an equation from page 2 and your answers from parts (i) and (ii) to calculate the current in the circuit.

[2]

current = A

- (iv) State the ammeter reading.

[1]

ammeter reading = A



- (b) Complete the following sentences by underlining the correct phrase or word. [3]

As the (**temperature** / **power** / **voltage**) of the thermistor is increased to 50 °C its resistance decreases. The total resistance in the circuit (**increases** / **stays the same** / **decreases**) and the circuit current (**increases** / **stays the same** / **decreases**).

- (c) (i) State how the original circuit could be changed to become a light sensing circuit. [1]

.....

- (ii) Give a reason for your answer. [1]

.....

